

Final Project GitHub Repository - The Hunter: Automated Email Phishing Defense Detection System

Course: ITAI 2376 - Deep Learning in AI

Institution: Houston City College

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Team Name: Team 1

Team Members:

- Chloe Tu
 - Matthew Choo
 - Franck Kolontchang
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Submission Details

- **GitHub Repository URL link:** [<https://github.com/Chloe-Tu2/The-Hunter-Automated-Email-Phishing-Defense-Detection-System>]
 - **Demo Video Link:** [Google Drive Demo Video](#)
 - **Agent Option:** Option B — Multi-Agent System (CrewAI · Three-Agent Sequential Pipeline)
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Final Project Submission: The Hunter

- **Chloe Tu** — Lead Developer (BiLSTM Model Training, CrewAI Orchestration, Gradio Dashboard)
 - **Matthew Choo** — Data Engineer (Dataset Preprocessing, SMOTE Balancing, Feature Engineering)
 - **Franck Kolontchang** — Systems Architect (Agent Design, Tool Integration, SQLite Threat Memory)
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Project Overview

The Hunter: Automated Email Phishing Defense Detection System

The Hunter is a fully autonomous, multi-agent AI pipeline that mimics a human Security Operations Center (SOC) to triage, classify, and verdict incoming emails for phishing threats in real time.

It addresses the gap in modern phishing defenses by deploying a team of three autonomous AI agents orchestrated through CrewAI. Each agent owns a distinct domain of expertise — perception (Email Ingest Specialist), risk assessment (Phishing Risk Analyst), and security response (SOC Orchestrator). The classification backbone is a stacked Bidirectional LSTM (BiLSTM) ensemble combined with a Logistic Regression feature model, evaluating threats and maintaining persistent threat memory via SQLite to automatically escalate repeat offenders.

Target User: Security Operations Centers (SOCs), IT Security teams at small-to-mid-size organizations, and cybersecurity students.

GitHub Repository Structure (Short Version)

```
The Hunter/
├── app/
│   ├── hunter_gradio_app.py    # Main Gradio web dashboard + CrewAI pipeline trigger
│   └── README.md               # app/ folder documentation
├── data/
│   ├── threat_memory.db        # SQLite threat memory database (auto-created at runtime)
│   └── README.md               # data/ folder documentation
├── docs/
│   ├── Final/
│   │   ├── FN_WriteUp_Tu_Chloe_Team_1_ITAI2376.md    # Final project writeup
│   │   └── FN_WriteUp_Tu_Chloe_Team_1_ITAI2376.pdf    # Final project writeup
│   │   (Markdown)
│   │   (PDF)
│   ├── Midterm/
│   │   └── MD_Blueprint_Tu_Chloe_Team_1_ITAI2376.md    # Midterm blueprint
│   │   (Markdown)
│   │   └── MD_Blueprint_Tu_Chloe_Team_1_ITAI2376.pdf    # Midterm blueprint (PDF)
│   │   └── The Hunter Architecture Diagram.png          # Full system architecture diagram
│   └── README.md                # docs/ folder documentation
├── models/
│   ├── bilstm_model.keras        # Trained BiLSTM neural network
│   ├── logreg_model.pkl          # Trained Logistic Regression feature model
│   ├── scaler.pkl                # StandardScaler for feature normalization
│   ├── threshold.pkl             # Calibrated precision-recall threshold
│   ├── tokenizer.pkl             # Keras tokenizer
│   └── README.md                 # models/ folder documentation
```

- notebooks/
 - Demo_The_Hunter_App.ipynb # Guided Colab demo notebook
 - The Hunter-Automated Email Phishing Defense...ipynb # Full training notebook
 - README.md # notebooks/ folder documentation
- results/
 - Pipeline_Execution.txt # Full 4-email pipeline terminal log
 - The_Hunter_crewai_Traces.txt # Curated CrewAI Crew Completion trace
 - README.md # results/ folder documentation
 - demo_visualizations/ # 22 screenshots from live Gradio
- dashboard runs
 - notebook_visualizations/ # 9 training charts exported from the
- notebook
 - .env.example # API key template
 - .gitignore # Git ignore file
 - LICENSE # MIT License
 - README.md # Full project documentation
 - REFLECTION.md # Team reflection
 - requirements.txt # Python dependencies
